Solution

A1.

$$\boxed{p_1 = \frac{mg}{S}}$$

A2. According to Pascal's law, the pressure of a liquid depends only on the depth of immersion, therefore

$$\boxed{p(z) = \rho g z}$$

B1. The equilibrium condition projected onto the vertical axis:

$$\boxed{\big(P(z+\mathrm{d}z) - P(z)\big)\pi R^2 - \mathrm{d}m\, g + \mu\sigma(z)\cdot 2\pi R\,\mathrm{d}z = 0}$$

B2. Using the relations $\mathrm{d}m = \rho\pi R^2\,\mathrm{d}z$ and $\sigma(z) = KP(z)$, we obtain

$$\frac{\mathrm{d}P}{\mathrm{d}z} + \frac{2K\mu P(z)}{R} - \rho g = 0$$

Solving the equation and taking into account that $P(0) = 0$, we obtain

$$P(z) = \frac{\rho g R}{2\mu K}\left(1 - e^{-2\mu K z/R}\right)$$

Let us find the pressure at the bottom by substituting $z = h$ into the expression obtained:

$$P(h) = \frac{\rho g R}{2\mu K}\left(1 - e^{-2\mu K h/R}\right)$$

Hence

$$\boxed{A = \frac{\rho g R}{2\mu K}, \qquad B = -\frac{\rho g R}{2\mu K}, \qquad \lambda = \frac{R}{2\mu K}}$$

B3. For small heights of the grain layer, the dependence of the pressure on the bottom on the height is approximately described by the expression

$$P(h) \approx \rho g h$$

Let us draw a tangent to the graph at the point $(0, 0)$. The slope of this tangent is

$$k_1 = 6.5\ \mathrm{kPa/m}$$

Hence

$$\boxed{\rho = \frac{k_1}{g} \approx 660\ \mathrm{kg/m^3}}$$

For sufficiently large h, the pressure on the bottom reaches a steady value. The steady-state pressure is

$$P_\infty = 26.7\ \text{kPa}$$

Hence

$$\boxed{K = \frac{\rho g R}{2\mu P_\infty} \approx 0.20}$$

C1.

$$\boxed{r = R - \alpha z}$$

C2. The equilibrium condition projected onto the vertical axis (here it is assumed that $\alpha \ll 1$, therefore $\cos\alpha \approx 1, \sin\alpha \approx \alpha$):

$$\boxed{P(z+\mathrm{d}z)\,\pi r^2(z+\mathrm{d}z) - P(z)\,\pi r^2(z) + 2\pi(\mu+\alpha)\sigma(z) r(z)\,\mathrm{d}z - \mathrm{d}m\,g = 0}$$

C3. Next, in the equation obtained in the previous step, we neglect the term α compared to μ. Note that near the free surface of the granular material the pressure is zero:

$$P(0) = A(1-\gamma) = 0$$

From this

$$\boxed{\gamma = 1}$$

The dependence $P(z)$ is described by the following differential equation:

$$\frac{\mathrm{d}(Pr^2)}{\mathrm{d}z} + 2K\mu P(z) r(z) - \rho g r^2(z) = 0$$

Let us introduce the notation $\xi = \dfrac{z}{z_{\max}}$. Substituting the expression for $P(z)$ given in the problem statement into the differential equation, taking into account that $\gamma = 1$:

$$\alpha^2 A z_{\max}\left(-1 + \beta(1-\xi)^{\beta-1}\right)(1-\xi)^2 + 2(K\mu-\alpha)A\alpha z_{\max}(1-\xi)\left(1-\xi-(1-\xi)^\beta\right) - \rho g \alpha^2 z_{\max}^2 (1-\xi)^2 = 0$$

The equation can be satisfied for all ξ only if the coefficients of $(1-\xi)^2$ and $(1-\xi)^{\beta+1}$ vanish. Therefore

$$-A\alpha^2 + 2(K\mu-\alpha)A\alpha - \rho g \alpha^2 z_{\max} = 0$$

Hence

$$\boxed{A = \frac{\rho g \alpha z_{\max}}{2K\mu - 3\alpha}}$$

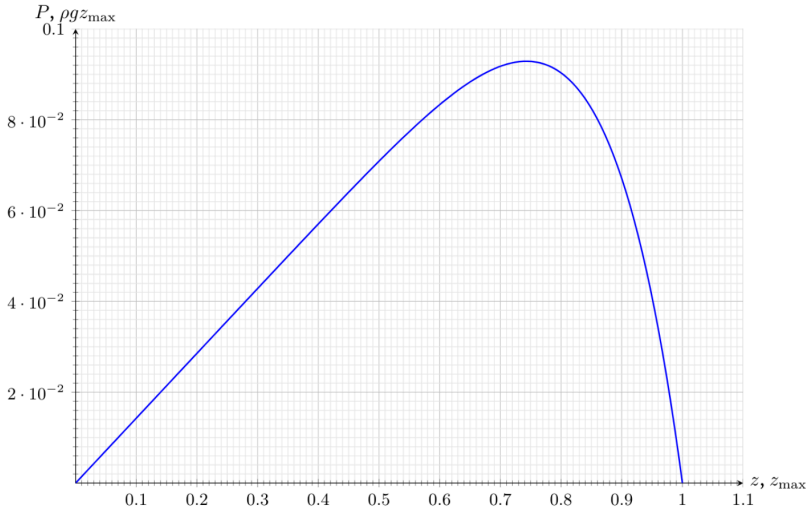
Also, the following holds:

$$\alpha^2 A\beta - 2(K\mu-\alpha)A\alpha = 0$$

From this

$$\boxed{\beta = \frac{2(K\mu - \alpha)}{\alpha}}$$

C4.



The pressure P (in units of $\rho g z_{\max}$) for $\alpha = 0.2K\mu$; the horizontal axis in units of $z_{\max}$.

Let us calculate for $\alpha = 0.2K\mu$:

$$A \approx 0.143 \rho g z_{\max}, \qquad \beta = 8$$

$P(z)$ is maximal when $1 - \xi - (1-\xi)^8 \to \max$, i.e.

$$8(1-\xi)^7 = 1, \qquad z = \left(1 - 8^{-\frac{1}{7}}\right) z_{\max} \approx 0.26 z_{\max}.$$

The corresponding pressure is

$$\boxed{P_{\max} \approx 0.093 \rho g z_{\max}}$$

> Editor's note: in the original graph the maximum lies at ≈ 0.74 on the horizontal axis, so that axis is in fact the distance $1 - z/z_{\max}$ measured from the apex. Against the depth z from the free surface, as in the calculation above, the maximum is at $z \approx 0.26 z_{\max}$ and the graph is mirrored: it rises steeply as $P \approx \rho g z$ near the free surface $z = 0$ and falls gently to zero at $z = z_{\max}$.

D1. The equilibrium condition projected onto the vertical axis (here the friction force acting on the granules from the vessel is directed downwards, since the force with which the granules act on the vessel lifts it):

$$\boxed{\big(P(z + \mathrm{d}z) - P(z)\big)\pi(R^2 - a^2) - \mathrm{d}m\, g - \mu_1 \sigma(z) \cdot 2\pi R\, \mathrm{d}z + \mu_2 \sigma(z) \cdot 2\pi a\, \mathrm{d}z = 0}$$

D2. The equation obtained has exactly the form of the equation of **B1**. Solving it, we obtain

$$\boxed{A = \frac{\rho g (R^2 - a^2)}{2(-\mu_1 R + \mu_2 a)K}, \qquad B = -\frac{\rho g (R^2 - a^2)}{2(-\mu_1 R + \mu_2 a)K}, \qquad \lambda = \frac{R^2 - a^2}{2(-\mu_1 R + \mu_2 a)K}}$$

D3. For a fixed height H of the granules, the dependence of the pressure $P(z)$ is likewise given by

$$P(z) = A + Be^{-z/\lambda}$$

Let us calculate the total force acting on the rod:

$$F = 2\pi aK\mu_2 \int_0^H P(z)\,\mathrm{d}z = 2\pi aK\mu_2 A\left(H - \lambda\left(1 - e^{-H/\lambda}\right)\right) \approx 2\pi aK\mu_2 A\frac{H^2}{2\lambda},$$

where we have used $H \ll \lambda$. By Newton's third law, a force of the same magnitude holds the granules up. In the critical situation this force must balance the gravity acting on the granules:

$$\rho g\pi(R^2 - a^2)H = \pi aK\mu_2 A\frac{H^2}{\lambda}$$

Expressing H from this,

$$\boxed{H = \frac{R^2 - a^2}{Ka\mu_2}}$$

E1. From the incompressibility condition,

$$\boxed{S_1\Delta l_1 = S_2\Delta l_2}$$

E2.

$$\boxed{W_1 = \rho S_1\Delta l_1\left(\frac{v_1^2}{2} + gh_1\right)}$$

> Editor's note: the original denotes this energy W_1 (the problem's E); the energy of the volume at the cross-section S_2 is written W_2 in the same way.

E3. At the pressure p_1 a volume V_1 of fluid is “pushed into” the stream tube, and at the pressure p_2 a volume $V_2 = V_1$ is “pushed out” of it. Therefore

$$\boxed{A = p_1V_1 - p_2V_2 = (p_1 - p_2)S_1\Delta l_1}$$

E4. We write the law of conservation of energy:

$$\rho S_1\Delta l_1\,\frac{v_2^2 + 2gh_2 - v_1^2 - 2gh_1}{2} = (p_1 - p_2)S_1\Delta l_1$$

Cancelling $S_1 \Delta l_1$ on both sides, we obtain

$$\boxed{p_2 = p_1 - \rho \frac{v_2^2 + 2gh_2 - v_1^2 - 2gh_1}{2}}$$

> Editor's note: the original answer (marking scheme included) lacks the factor ρ, which the equation just above shows must be there.

E5. From the Bernoulli equation derived above, for the flow of an ideal fluid

$$p + \rho g h + \rho \frac{v^2}{2} = \text{const}$$

The pressure at the hole equals the atmospheric pressure, and so does the pressure at the upper boundary between the fluid and the air. Therefore

$$p_0 + \rho g h + 0 = p_0 + 0 + \rho \frac{v(h)^2}{2}$$

and hence

$$\boxed{v(h) = \sqrt{2gh}}$$

E6. The volume flow rate through the hole is

$$\dot{V} = -\frac{\pi D^2}{4} \sqrt{2gh}$$

The continuity equation gives the differential equation for the water height above the hole:

$$\pi R^2 \dot{h} = -\frac{\pi D^2}{4} \sqrt{2gh}$$

Since $R \gg D$, $\dot{h} \ll \sqrt{2gh}$, so the assumption that the fluid velocity far from the hole can be neglected is justified. Now the outflow time can be expressed:

$$\boxed{\tau = \frac{2\sqrt{2}R^2}{D^2\sqrt{g}} \int_0^H \frac{\mathrm{d}h}{\sqrt{h}} = \frac{4\sqrt{2}R^2\sqrt{H}}{D^2\sqrt{g}}}$$

F1. In the critical situation the surface of the granular material is inclined at the angle θ_r to the horizontal and touches the outer edge of the hole. From this,

$$\boxed{\omega_{\max} = D \cot \theta_r}$$

F2. Let us estimate the velocity of the granules at the exit of the hole using the law of conservation of energy:

$$v \propto \sqrt{D}$$

Let us estimate the mass flow rate of the granules:

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \rho S v \propto D^2 \sqrt{D} \propto D^{5/2}$$

From this

$$\boxed{n = \frac{5}{2}, \qquad q = 0}$$

Solution

A0.

$$\boxed{d_e = X_0 \ln 2 = 25.1\ \mathrm{g/cm^2}}$$

A1. The energy of the shower particles halves after every splitting length, so let us find after how many splittings the energy of a particle falls below the critical energy:

$$\log_2 \frac{E_\gamma}{E_e^{\mathrm{crit}}} = 6.8 \ \Rightarrow$$

$$\boxed{n_{\max} = 7}$$

A2. Let us calculate the radiation path length the shower has passed by the 7th splitting:

$$x_e = d_e \cdot n_{\max} = 175.7\ \mathrm{g/cm^2}.$$

In an atmosphere of constant density ρ_0 this radiation path length corresponds to the geometric distance

$$\boxed{l_e = \frac{x_e}{\rho_0} = 1.4\ \mathrm{km}}$$

A3. Every splitting doubles the number of shower particles. Therefore

$$\boxed{N_{\max} = 2^{n_{\max}} = 128}$$

A4. At every splitting a photon turns into an electron–positron pair and disappears, while the electron and the positron each emit one photon. Hence the number of photons after a splitting equals the sum of the numbers of electrons and positrons before it, and the numbers of positrons and electrons each grow by the number of photons before the splitting. Let us tabulate the number of particles of each kind by generation:

n	0	1	2	3	4	5	6	7
γ	1	0	2	2	6	10	22	42
e^-	0	1	1	3	5	11	21	43
e^+	0	1	1	3	5	11	21	43

When the development of the shower stops there are $n_{e^-} = 43$ electrons in it. The total charge carried by the electrons is therefore

$$\boxed{Q_e = n_{e^-} \cdot q_e = -6.9 \cdot 10^{-18}\ \mathrm{C}}$$

B1. The number of generations of the hadronic component is found in the same way as in **A1**, except that in a hadronic shower the energy of the particles drops 15 times at every splitting:

$$\log_{15} \frac{E_p}{E_\pi^{\text{crit}}} = 2.2 \;\Rightarrow$$

$$\boxed{n_{\text{hadr}}^{\max} = 3}$$

B2. In a hadronic interaction the energy of the particles drops 15 times, whereas in an electromagnetic one it drops 2 times. The number of generations is therefore largest when, apart from the first hadronic interaction (the proton breaks up and a π^0 meson splits into two photons), the shower develops electromagnetically from the start:

$$\log_2 \frac{E_p/(15 \cdot 2)}{E_e^{\text{crit}}} = 11.4 \;\Rightarrow$$

$$\boxed{n_{\text{EM}}^{\max} = 12 + 1 = 13}$$

B3. The energy carried away by muons and neutrinos is the energy left in the charged pions $\pi^\pm$ when their energy falls below the critical value. In every hadronic interaction 6 of the 15 particles are neutral pions π^0, whose energy goes to the electromagnetic component; the other 9 of the 15 stay in the hadronic component.

The fraction of the energy left in the hadronic component when the hadronic shower stops developing (after $n_{\text{hadr}}^{\max} = 3$ splittings) is therefore

$$\beta_{\text{miss}} = \left(\frac{9}{15}\right)^{n_{\text{hadr}}^{\max}} = \left(\frac{3}{5}\right)^3 = 0.216.$$

All these charged pions later decay into muons and neutrinos, which carry this energy away. Therefore

$$\boxed{\beta_{\text{miss}} = 0.22}$$

C1. One hadronic interaction reduces the energy of a particle 15 times, which corresponds to $\log_2 15 = 3.9$ electromagnetic splittings. Reducing the energy 15 times takes the radiation path length $d_h = 57\ \text{g/cm}^2$, or $3.9 \cdot d_e = 97.9\ \text{g/cm}^2$. The electromagnetic shower therefore stops developing later (farther away), so h_{stop} must be calculated for the electromagnetic component.

Let us calculate the radiation path length passed until the shower stops: one hadronic interaction, then the decay of a π^0 into two photons, and the remaining electromagnetic splittings:

$$\log_2 \frac{E/(15 \cdot 2)}{E_e^{\text{crit}}} = 15.2 \;\Rightarrow\; x = d_h + 16 \cdot d_e = 458.6\ \text{g/cm}^2.$$

Now, taking the density of the atmosphere into account, we calculate the geometric altitude at which the shower stops developing:

$$x = \int_{h_{\text{stop}}}^{\infty} \rho_0 e^{-h/H}\, dh = \rho_0 H e^{-h_{\text{stop}}/H}.$$

$$\boxed{h_{\text{stop}} = H \ln \frac{\rho_0 H}{x} = 6.6 \text{ km}}$$

C2. As in **B3**, let us find the energy carried away by muons and neutrinos:

$$\log_{15} \frac{E}{E_\pi^{\text{crit}}} = 3.14 \;\Rightarrow\; n_h = 4.$$

All the remaining energy then stays in the electromagnetic component:

$$\boxed{\beta_{\text{EM}} = 1 - \left(\frac{9}{15}\right)^4 = 0.87}$$

C3. Let us calculate the air density at the altitude h_{stop}:

$$\rho(h_{\text{stop}}) = \rho_0 e^{-h_{\text{stop}}/H} = 0.57 \text{ kg/m}^3.$$

Substituting this value into the formula for the radius, we obtain

$$\boxed{R_{\text{stop}} = \frac{X_0}{\rho_0 e^{-h_{\text{stop}}/H}} \cdot \frac{E_s}{E_e^{\text{crit}}} = 150 \text{ m}}$$

C4. Converting the shower radius down to the ground, taking the change of the air density into account, we obtain

$$R = R_{\text{stop}} \frac{\rho}{\rho_0} = 67 \text{ m}.$$

$$\boxed{S = \pi R^2 = 14 \cdot 10^3 \text{ m}^2}$$

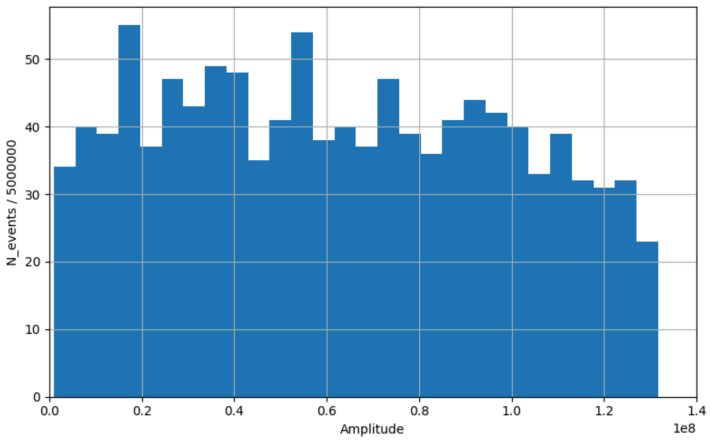
C5. The particles diverge at the altitude h_{stop} at the angle $\theta = R_{\text{stop}}/h_{\text{stop}}$, so when the shower front reaches the ground there is a path difference

$$\Delta = h_{\text{stop}} \cdot (1 - \cos\theta) \approx h_{\text{stop}} \frac{\theta^2}{2} = \frac{R_{\text{stop}}^2}{2h_{\text{stop}}} = 1.7 \text{ m}.$$

The particles move almost at the speed of light, so

$$\boxed{\Delta t = \frac{\Delta}{c} = \frac{R_{\text{stop}}^2}{2ch_{\text{stop}}} = 5.7 \text{ ns}}$$

D1. To find κ from the data, let us plot the histogram of the detector signals.



Histogram of the detector signal amplitudes.

The energy of a particle reaching the detector cannot exceed the critical energy $E_e^{\text{crit}} = 88$ MeV. The histogram shows a sharp cutoff at the signal value $A^{\text{crit}} = 1.32 \cdot 10^8$, from which

$$\boxed{\kappa = \frac{A^{\text{crit}}}{E_e^{\text{crit}}} = 1.5 \cdot 10^9 \text{ GeV}^{-1}}$$

D2. Let us add up all the signals recorded by the detectors:

$$A_0 = 70240347424,$$

from which, with the coefficient κ,

$$\boxed{E_0 = \frac{A_0}{\kappa} = 46.8 \text{ GeV}}$$

D3. Using the formula of **C3**, let us calculate the mean shower radius from the data:

$$R = \sum_i \frac{\sqrt{x_i^2 + y_i^2}}{N} = 68.3 \text{ m}.$$

When the shower stopped developing, its radius was, according to formula (8),

$$R_{\text{stop}} = \frac{X_0}{\rho(h_{\text{stop}})} \cdot \frac{E_s}{E_e^{\text{crit}}},$$

and then, according to formula (9), it shrank in inverse proportion to the density:

$$R = R_{\text{stop}} \frac{\rho(h_{\text{stop}})}{\rho_{\text{atm}}} = \frac{X_0}{\rho_{\text{atm}}} \cdot \frac{E_s}{E_e^{\text{crit}}}.$$

$$\boxed{\rho_{\text{atm}} = \frac{X_0}{R} \cdot \frac{E_s}{E_e^{\text{crit}}} = 1.26 \text{ kg/m}^3}$$

Solution

A1. Let A be the contact area. The volume charge density is $\rho = \pm q_e N_{a,d}$ ($+$ for holes, $-$ for electrons), so the condition that the total charge of the holes and electrons is zero means $AN_a d_p = AN_d d_n$. Therefore

$$\boxed{N_a d_p = N_d d_n.}$$

A2. By Gauss's law in differential form,

$$\operatorname{div}\vec{E} = \frac{\rho}{\varepsilon_0 \varepsilon}.$$

Since $\vec{E} = \vec{E}(x)$,

$$\operatorname{div}\vec{E} = \frac{dE_x}{dx},$$

which proves the given formula.

A3. Using the result of A2 together with the relation between the concentration and the volume charge density, $\rho = \pm q_e N_{a,d}$ ($+$ for holes, $-$ for electrons), we obtain

$$\frac{dE_x}{dx} = \begin{cases} -\dfrac{q_e N_a}{\varepsilon_0 \varepsilon}, & -d_p \le x \le 0, \\ \dfrac{q_e N_d}{\varepsilon_0 \varepsilon}, & 0 < x \le d_n, \\ 0, & x < -d_p \text{ or } x > d_n. \end{cases}$$

Now we need the indefinite integral and the constant of integration. To get the answer faster, let us compute the definite integral directly. Since $E_x(\pm\infty) = 0$,

$$E_x(x) = \int_{-\infty}^{x} E'(x)\,dx = -\int_{x}^{+\infty} E'(x)\,dx.$$

Using these relations, we obtain the final answer:

$$\boxed{E_x(x) = \begin{cases} -\dfrac{q_e N_a (x + d_p)}{\varepsilon_0 \varepsilon}, & -d_p \le x \le 0, \\ \dfrac{q_e N_d (x - d_n)}{\varepsilon_0 \varepsilon}, & 0 < x \le d_n, \\ 0, & x < -d_p \text{ or } x > d_n. \end{cases}}$$

A4. In one dimension $\varphi'(x) = -E_x(x)$. With the condition $\varphi(0) = 0$, the electrostatic potential can be written as

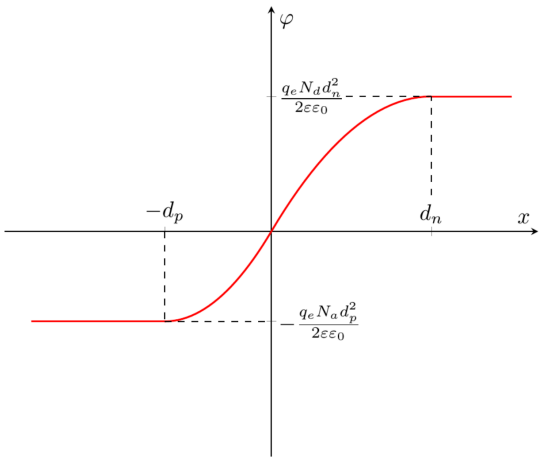
$$\varphi(x) = -\int_0^x E_x(x)\,\mathrm{d}x.$$

Computing the integral directly, we obtain

$$\boxed{\varphi(x) = \begin{cases} -\dfrac{q_e N_a d_p^2}{2\varepsilon_0 \varepsilon}, & x < -d_p, \\ \dfrac{q_e N_a (x^2 + 2d_p x)}{2\varepsilon_0 \varepsilon}, & -d_p \le x < 0, \\ \dfrac{q_e N_d (2d_n x - x^2)}{2\varepsilon_0 \varepsilon}, & 0 \le x < d_n, \\ \dfrac{q_e N_d d_n^2}{2\varepsilon_0 \varepsilon}, & x \ge d_n. \end{cases}}$$

$$\boxed{\Delta\varphi = \frac{q_e \left(N_a d_p^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0}.}$$

A5.



A qualitative graph of the potential $\varphi(x)$.

A6. From A4,

$$\Delta\varphi = \frac{q_e \left(N_a d_p^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0}.$$

Using the relation $N_a d_p = N_d d_n$ obtained in A1,

$$\Delta\varphi = \frac{q_e \left(N_a d_p^2 + \frac{N_a^2}{N_d} d_p^2\right)}{2\varepsilon\varepsilon_0} = \frac{q_e N_a (N_a + N_d) d_p^2}{2\varepsilon\varepsilon_0 N_d}.$$

Similarly,

$$\Delta\varphi = \frac{q_e \left(\frac{N_d^2}{N_a} d_n^2 + N_d d_n^2\right)}{2\varepsilon\varepsilon_0} = \frac{q_e N_d (N_a + N_d) d_n^2}{2\varepsilon\varepsilon_0 N_a},$$

which gives the answer:

$$\boxed{d_p = \sqrt{\frac{2\varepsilon\varepsilon_0 \Delta\varphi N_d}{q_e N_a (N_a + N_d)}} \approx 58.7\ \text{nm}, \qquad d_n = \sqrt{\frac{2\varepsilon\varepsilon_0 \Delta\varphi N_a}{q_e N_d (N_a + N_d)}} \approx 117.5\ \text{nm}.}$$

A7. From $E_x(x)$ obtained in A3, the absolute value of the electric field is maximal at $x = 0$, where

$$E_{\max} = \frac{q_e N_a d_p}{\varepsilon \varepsilon_0}.$$

Substituting the expression for d_p from A6, we obtain

$$\boxed{E_{\max} = \sqrt{\frac{2 q_e \Delta\varphi N_a N_d}{\varepsilon \varepsilon_0 (N_a + N_d)}} \approx 9.08 \cdot 10^6 \text{ V/m}.}$$

B1. From the distribution of $n(x)$,

$$\frac{n(x = +\infty)}{n(x = -\infty)} = \exp\left(-\frac{U_n(x = +\infty) - U_n(x = -\infty)}{kT}\right)$$

Before the voltage is applied, $n(x = +\infty) = N_d$ and $n(x = -\infty) = n_{10}$. When the voltage is applied, the potential energy difference of an electron $U_n(x = +\infty) - U_n(x = -\infty)$ changes by $(-q_e) \cdot (-V) = q_e V$. According to the problem statement, the electron concentration in the n-type semiconductor is N_d regardless of the voltage. Hence after the voltage is applied still $n(x = +\infty) = N_d$, but $n(x = -\infty) = n_1$. Therefore

$$\frac{N_d}{n_1} = \exp\left(-\frac{q_e V}{kT}\right) \cdot \frac{N_d}{n_{10}},$$

and hence

$$n_1 = n_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

The expression for p_1 is obtained in the same way. From the distribution of $p(x)$,

$$\frac{p(x = +\infty)}{p(x = -\infty)} = \exp\left(-\frac{U_p(x = +\infty) - U_p(x = -\infty)}{kT}\right).$$

Before the voltage is applied, $p(x = +\infty) = p_{10}$ and $p(x = -\infty) = N_a$. When the voltage is applied, the potential energy difference of a hole $U_p(x = +\infty) - U_p(x = -\infty)$ changes by $q_e \cdot (-V) = -q_e V$. According to the problem statement the hole concentration in the p-type semiconductor is N_a, so after the voltage is applied still $p(x = -\infty) = N_a$, but $p(x = +\infty) = p_1$. Therefore

$$\frac{p_1}{N_a} = \exp\left(\frac{q_e V}{kT}\right) \cdot \frac{p_{10}}{N_a},$$

and hence

$$p_1 = p_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).$$

$$\boxed{n_1 = n_{10} \cdot \exp\left(\frac{q_e V}{kT}\right), \qquad p_1 = p_{10} \cdot \exp\left(\frac{q_e V}{kT}\right).}$$

B2. Let A be the cross-sectional area of the n-type semiconductor. Let us calculate the change of the number of holes in the small volume considered during a short time Δt. The change due to recombination is $\left(\frac{dp_1}{dt}\right)_{\text{rec}} A\Delta x\Delta t$. The changes due to diffusion through the two planes are $J_p(x)A\Delta t$ (at the coordinate x) and $-J_p(x+\Delta x)A\Delta t$ (at $x+\Delta x$). The total number of holes, and hence their concentration, is constant, so the total change must be zero:

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} A\Delta x\Delta t + J_p(x)A\Delta t - J_p(x+\Delta x)A\Delta t = 0.$$

Therefore

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = \frac{J_p(x+\Delta x) - J_p(x)}{\Delta x} = \frac{dJ_p(x)}{dx} = -D_p\frac{d^2p_1}{dx^2}.$$

Δp and p_1 differ only by a constant, so

$$\frac{d^2p_1}{dx^2} = \frac{d^2\Delta p}{dx^2}.$$

Taking into account that

$$\left(\frac{dp_1}{dt}\right)_{\text{rec}} = -\frac{\Delta p}{\tau_p},$$

we obtain

$$\boxed{\frac{d^2\Delta p}{dx^2} = \frac{\Delta p}{\tau_p D_p}}$$

B3. The general solution of the equation obtained in B2 has the form

$$\Delta p = a\exp\left(-\frac{x}{\sqrt{\tau_p D_p}}\right) + b\exp\left(\frac{x}{\sqrt{\tau_p D_p}}\right),$$

where a and b are coefficients. As $x \to +\infty$ the hole concentration must approach the equilibrium value p_{10}, so $\Delta p \to 0$ and $b = 0$. Therefore

$$\Delta p \sim \exp\left(-\frac{x}{\sqrt{\tau_p D_p}}\right),$$

and Δp changes e times when x changes by

$$\boxed{L_p = \sqrt{\tau_p D_p} \approx 2.24 \cdot 10^{-5}\ \text{m}.}$$

B4. From the expression for Δp obtained in B3, for $x > d_n$

$$p_1(x) = p_{10} + a\exp\left(-\frac{x}{L_p}\right),$$

where a is some coefficient. Inside the depletion layer the concentration $p(x)$ follows the Boltzmann distribution. Therefore

$$\frac{p(d_n)}{p(-d_p)} = \exp\left(-\frac{U_p(d_n) - U_p(-d_p)}{kT}\right).$$

According to the problem statement the electric field outside the depletion layer can be taken as zero, so the whole voltage drop occurs within $-d_p \le x \le d_n$. Hence, when the voltage is applied, the potential energy difference of a hole $U_p(d_n) - U_p(-d_p)$ changes by $-q_eV$, and as in B1 we can write

$$\frac{p_1(d_n)}{N_a} = \exp\left(\frac{q_eV}{kT}\right)\frac{p_{10}}{N_a},$$

from which

$$p_1(d_n) = \exp\left(\frac{q_eV}{kT}\right)p_{10} = p_{10} + a\exp\left(-\frac{d_n}{L_p}\right),$$

so that

$$a = p_{10}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(\frac{d_n}{L_p}\right).$$

$$\boxed{p_1(x) = p_{10} + p_{10}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(-\frac{x - d_n}{L_p}\right)}$$

B5. To calculate the hole current, consider the n-type region near the boundary of the depletion layer, where the hole flux is determined by diffusion:

$$\begin{aligned} J_p = -D_p\frac{dp_1}{dx} &= \frac{D_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\exp\left(-\frac{x - d_n}{L_p}\right)\Bigg|_{x=d_n} \\ &= \frac{D_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right). \end{aligned}$$

The current density is $j_p = q_eJ_p$, so

$$\boxed{j_p = \frac{q_eD_pp_{10}}{L_p}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).}$$

B6. As in B5, the electron current has the form

$$j_n = \frac{q_eD_nn_{10}}{L_n}\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right).$$

Taking into account that

$$p_{10} = \frac{n_i^2}{N_d}, \qquad n_{10} = \frac{n_i^2}{N_a},$$

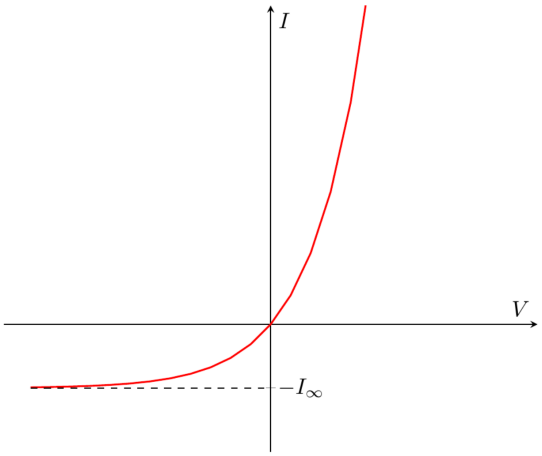
the required current, the sum of the hole and electron currents, is

$$I = q_eAn_i^2\left(\exp\left(\frac{q_eV}{kT}\right) - 1\right)\left(\frac{D_p}{L_pN_d} + \frac{D_n}{L_nN_a}\right).$$

Using the answer of B3, we obtain

$$\boxed{I = q_e A n_i^2 \left(\exp\left(\frac{q_e V}{kT}\right) - 1\right)\left(\frac{1}{N_d}\sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_a}\sqrt{\frac{D_n}{\tau_n}}\right).}$$

B7.



A qualitative graph of $I(V)$.

B8. The saturation current is

$$I_\infty = q_e A n_i^2 \left(\frac{1}{N_d}\sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_a}\sqrt{\frac{D_n}{\tau_n}}\right) = 5.77 \cdot 10^{-16}\ \text{A},$$

and the current at $V = 0.7$ V is $I_1 = 3.23 \cdot 10^{-4}$ A.

$$\boxed{I_\infty = 5.77 \cdot 10^{-16}\ \text{A}, \qquad I_1 = 3.23 \cdot 10^{-4}\ \text{A}.}$$

> Editor's note: the numbers of the original (marking scheme included) correspond to $n_i = 1.5 \cdot 10^{16}\ \text{m}^{-3}$. With the $n_i = 1.0 \cdot 10^{16}\ \text{m}^{-3}$ given in the problem, $I_\infty \approx 2.56 \cdot 10^{-16}$ A and $I_1 \approx 1.44 \cdot 10^{-4}$ A.